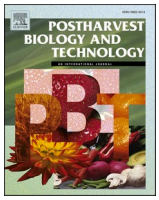
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Development and evaluation of an apple infield grading and sorting system

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ABSTRACT

Infield pre-sorting is intended to remove processing-grade (inferior) fruit that are not suitable for the fresh market, so that growers can handle sorted apples differentially in postharvest storage and packing to achieve cost savings. To achieve this goal, we developed an apple infield grading and sorting system, in which fruit singulation, rotation, and transportation are achieved by using simple, compact pitch-variable screw conveyors, fruit grading (for size and color) is accomplished with a low-cost imaging system, and fruit sorting is done by using paddle sorters. Experiments were conducted for 'Red Delicious' and 'Golden Delicious' apples to evaluate the overall performance of the infield grading and sorting system in terms of grading repeatability (i.e., chances of each apple that would be graded into the same quality grade in multiple runs and different lanes), bruising damage, and sorting accuracy (consistency between the imaging-based grading results and destinations). Results showed that the grading repeatability rates of the system were above 90 % and 81 % for intra- and inter-lane grading, respectively. The system achieved above 99 % sorting accuracy for the system throughputs of 7.5, 9.0, and 10.5 fruit s^{-1}, while 100 % of sorted apples were graded Extra Fancy and 55 % or higher of the apples incurred no bruising damage during the grading and sorting process. The infield grading and sorting system is compact and robust in performance, and it can meet commercial infield sorting needs.

1. Introduction

Currently in the United States, freshly harvested apples of different quality grades are immediately placed in the bins at orchards and then hauled to storage facilities for either short-term refrigerated air (RA) or long-term controlled atmosphere (CA) storage (Zhang et al., 2017). In Michigan, which is one of the major apple producing states in the U.S., it was reported to cost about \$30 t^{-1} $month^{-1}$ for RA storage and about \$80 t^{-1} $month^{-1}$ for CA storage (personal communication with Riveridge Packing LLC, and Elite Apple Co. LLC, MI, USA). After being removed from storage, these apples need to go through the packing line for grading, sorting, and packaging. High quality fruit are destined for the fresh market, while low quality or defective fruit (processing grade) are sent for making juice and/or processed products (e.g., apple sauce, fresh cut slices, and canned sliced apples). Under the current practice, growers would pay the same rate for postharvest storage and packing of all harvested fruit, regardless of quality grades (Mizushima and Lu, 2011b; Zhang et al., 2017). If there is a high percentage of low-quality apples in the batches, growers may not break even (Schotzko and Granatstein, 2005) due to low revenue but high storage and packing cost for processing-grade apples. Hence, infield presorting of harvested fruit into two quality grades (i.e., processing and fresh) could help growers achieve significant cost savings by implementing more cost-effective postharvest handling procedures. For instance, processing-grade apples can be kept in RA, instead of CA, storage, or directly sent to the processor, so as to reduce or even eliminate the cost of postharvest storage and packing. A case study by Zhang et al. (2017) showed that for 10 t apples that are composed of 60 % and 40 % fresh and processing-grade apples respectively, infield sorting (assuming 90 % sorting accuracy) would save about 35 % of the total postharvest storage and packing cost. Hence, there is a significant commercial potential for adopting infield sorting practices (Mizushima and Lu, 2011b).

One potential solution to infield sorting is to manually grade and sort apples during harvest and place them in different bins according to quality grades. However, growers are generally reluctant to adopt the manual sorting method because it would mean additional labor and lower harvest productivity. Workers also do not have incentives to perform infield sorting because they are usually paid based on their fruit

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picking rate. In addition, manual sorting is challenging to implement because workers are prone to visual variability or inaccuracy and fatigue in quality grading of apples (Zhang, 2015; Lu et al., 2017). An alternative and potentially more feasible solution is the deployment of machine-vision based sorting technology in orchards. Despite rapid progress in sensing and automation technologies for agricultural applications over the past decades (Zhang et al., 2018, 2020; Fu et al., 2020a, 2020b), automated infield grading and sorting practices have so far not been adopted for commercial harvest of apples, mainly due to the lack of appropriate technologies meeting the requirements of low equipment cost, high throughput, reliability, and robustness.

Successful fruit sorting requires segregation and accurate grading of apples first. If apples are not spaced apart and arranged in an orderly manner, the machine vision-based grading system would have difficulty in accurately segregating the fruit from the acquired images, especially when they overlap or adjoin, which could result in inaccurate grading results and also complicate subsequent sorting of the graded fruit (Pothula et al., 2018). Hence, individual apples need be separated and arranged in a specific format before images are taken for grading. An extensively adopted separation method is to transport apples via multiple belts with different speeds (Rohrbach and McClure, 1978). When shifting from low- to high-speed belts, apples are physically separated due to the acceleration. Another approach applies liquid medium for fruit conveyance and uses a V-shaped conveyor and diabolo-shaped element (i.e., two ends are higher while the center is low) for fruit separation (De Vos and Van Arkel, 2003). Additionally, Agrosaw (2019) fed apples into a spinning disk and took advantage of centrifugal force for singulation. However, all these existing fruit singulation technologies are primarily designed for packinghouse use; they are bulky or large in size and too complex and expensive for infield use.

Following or accompanied with the singulation process, apples need be in rotation status so that the imaging system can take multiple images from each fruit to cover the whole surface at least once in order to achieve accurate quality grading. Bi-cone rollers are commercially used in packinghouses to rotate individual fruit by friction (Regier and Hiebert, 1993; Hiebert, 1997; Throop et al., 2001). Throop et al. (2003) designed a fruit rotation mechanism, consisting of a ring and a wheel protruding up through the center of the ring. The ring holds an apple loosely and the rotating wheel drives the apple rotation when contacting the fruit cheek (Keesling, 1965; Tichy, 1988; Ross and Meissner, 1996). Blasco et al. (2003) developed a movable suction cup to hold and then rotate an apple, so that the camera could inspect the majority of the apple's surface (the apple surface attached to the suction cup is blocked). However, the rotating wheel method is complicated in design and would require large space, and it would be also prone to malfunctioning in the rugged orchard environment. The suction cup design, on the other hand, could not meet the high throughput requirement (i.e., $>$ 9 apples s^{-1}). Our group [i.e., the research team with the U.S. Department of Agriculture Agricultural Research Service (USDA/ARS) at East Lansing, Michigan] previously developed a laboratory bi-cone roller system for imaging apples (Mizushima and Lu, 2011a), but the system is not suitable for infield use, because it is complicated in the mechanical design and would need a separate mechanism to singulate fruit.

By real-time collecting and processing multiple images that cover the whole surface of an apple, the machine vision system can objectively assess the fruit based on color and size and then assign a quality grade to the inspected fruit according to the established grading standards or the user's specific requirements (Cubero et al., 2014; Pothula et al., 2018; Zhang et al., 2018). For easy transportation (i.e., height restrictions) and convenient infield use, the imaging chamber needs be compact with a limited height (Zhang et al., 2019). In previous research, our group developed a compact imaging chamber with an overall height of 610 mm by using a camera with a large field of view (FoV) (Mizushima and Lu, 2013a). The image distortion issue associated with the large FoV camera is addressed by adoption of a variable-pixel per unit dimension method (Mizushima and Lu, 2011a). Algorithms were developed for fruit grading based on color and size, which demonstrated high accuracy and reliability in laboratory tests at a throughput of lower than 6 fruit s^{-1} (Mizushima and Lu, 2013a). However, the performance (or repeatability) of the grading system has not been tested, especially at high throughputs. Repeatability herein refers to the grading results of the same apples by the machine vision system in multiple runs, and it is critical to have consistent grading results for multiple runs.

After apples have been graded, the sorting system would then direct them into different destinations (or bins) according to the grading results. Several groups of researchers have reported an air jet principle for apple sorting, in which an air ejector is triggered to push apples to its destination (Rohrbach and McClure, 1978; Delwiche et al., 1993; Blasco et al., 2009; Furniss, 2011). However, the rugged field conditions (e.g., machine vibration and uneven terrains) would pose a challenge for adoption of the air jet principle. Another type of sorting system is to hold and track apples in individual cups, and when an apple arrives at its destination, the cup is triggered to tilt so that the fruit exits the cup to its destination (Anonymous, 2016; Sofu et al., 2016). In earlier research, our group developed a new cup-based sorting system, with which graded apples are held in the cups and drop to the target bins from the lower side of the conveyor (Pothula et al., 2018). This special cup-dropping design reduces the vertical dimension of the sorting system significantly, compared to the conventional cup-based sorting system that drops apples from the cups that are still moving on the upper side of the conveying system. While the sorting system performed satisfactorily (Pothula et al., 2018), it needs specially designed cups (thus higher cost) and was found to be less reliable or prone to malfunctioning, when it ran at high sorting throughputs (i.e., $>$ 6 apples s^{-1}). In further research to simplify the fruit sorting mechanism, our group designed and fabricated a new version of rotary sorter (Lu et al., 2018). The sorter consists of rotating disks with four compartments. As the sorter is rotating in synchronization with the fruit singulation/rotation mechanism, the graded apples, after leaving the imaging chamber, fall into the individual compartments of the rotary sorter, which are being sent to the destination bins via computer-controlled paddles. Compared to the cup-based sorters, the rotary sorter is simpler and more compact in design; it can sort apples into three quality grades (i.e., cull, processing and fresh). However, laboratory and field tests showed that when the sorter was running at a throughput higher than 6 fruit s^{-1}, its performance decreased with increased bruising damage to apples (unpublished) and lowered sorting accuracy. To fully meet the harvest productivity of a harvest crew of 6 workers (a skilled worker can pick one fruit per second), the grading and sorting system should be able to handle at least 6 harvested fruit per second, while causing no or minimum bruising to the fruit.

This paper reports the latest progress in our continued effort on the development of a new apple grading and sorting system towards commercial infield presorting applications, with the following specific objectives: 1) design a new paddle-based sorter for integration with the newly developed apple singulation, rotation and transportation mechanism, 2) evaluate the intra- and inter-lane grading repeatability of the machine vision-based grading system at high throughputs of 7.5–10.5 fruit s^{-1}, and 3) evaluate the apple sorting accuracy of the new sorter at different throughputs and the quality grades of apples due to bruise damage incurring in grading and sorting.

2. Materials and methods

2.1. Grading system

The apple grading system consists of two main parts: an apple singulation, rotation and transportation mechanism (SRTM) and an imaging-based grading system. The SRTM is responsible for singulating, rotating and transporting the fruit and presenting them in an individual manner to the imaging-based grading system, which acquires and processes images and provides grading results for individual apples in real

time.

2.1.1. Apple singulation, rotation and transportation mechanism

Overall system size, simplicity and reliability were key considerations in the design of an infield fruit grading system. It was, therefore, desirable and necessary for fruit singulation, rotation and transportation functions to be accomplished using a single and compact mechanism, instead of multiple ones described above. Our group developed the new pitch-variable screw conveyors, which simultaneously perform the three functions of fruit singulation, rotation, and transportation (Fig. 1). There are three sets (lanes) of pitch-variable screw conveyors for the current design, each being made up of a pair of aluminum tubes (32 mm diameter) covered with soft foam (13 mm thickness), and the screw thread is made of multi-purpose neoprene foam strips (16 mm width and 9 mm height) (McMaster-Carr, Aurora, OH, USA), whose pitch (designated as D1, D2, and D3 in Fig. 1B) increases from 83 mm to 121 mm from the fruit entry side to the exit side of the conveyors (1560 mm in total length). As harvested apples enter the three rotating screw conveyors (synchronized in the same direction) in a random pattern (clustered or separate) from harvest conveyors, these apples are being dispersed gradually, owing to the increasing pitches of the screw conveyors along the fruit moving direction, until they are completely separated into tandem arrangements. As the pairs of screw conveyors are rotating in the same direction, the friction between the fruit and the soft foam surface of the conveyors enables the apples to rotate in reverse direction with respect to the rotating conveyors. By the time when the apples approach the final section of the screw conveyor (i.e., the D3 section of the screw conveyor in Fig. 1), they are completely separated and ready for imaging.

This system was designed for incorporation into a self-propelled harvest-assist platform that can accommodate a harvest crew of 6 workers (Lu et al., 2021). Since a skilled worker can pick approximately one apple in a second, the system throughput was designed to run at the full capacity of 10.5 apples s^{-1}, so that it has extra capacity to handle unexpected surges in fruit throughput due to the overall uneven picking rates by the harvest crew, thus avoiding potential fruit accumulations at the entrance of the screw conveyors. The pitch-variable screw conveyor sets were experimentally demonstrated to be capable of handling more than 15 apples s^{-1} in the three-lane configuration (i.e., 5 apples s^{-1} $lane^{-1}$), which is more than enough to meet the full throughput requirement of 10.5 apples s^{-1}.

2.1.2. Imaging-based fruit grading

As apples are entering the final section of the screw conveyors, above which an imaging chamber is installed, they have already been singulated and aligned in three lanes. Mounted on the top of the imaging chamber is a low-cost CCD color camera (Fire I Digital Camera, Unibrain Inc., San Ramon, CA, USA), which faces downward vertically and covers the three lanes of screw conveyors with a FoV area of approximately 460 mm × 340 mm (L × W) and a vertical distance of approximately 600 mm from the screw conveyors. Four pairs of two 14 W white LED lamps with a length of 609 mm (Forest Lighting Inc., Marietta, GA, USA) are mounted above and along the three lanes of screw conveyors to provide uniform illumination for apples entering the imaging chamber. Graphical user interface (GUI) software was developed in the environment of Qt 5.10 to control the vision and sorting system, in which the library OpenCV 3.4.5 was deployed to perform image processing.

The vision system grades the apples into different quality grades based on fruit size and color attributes. In this study, the apples were graded into two quality grades, i.e., fresh and processing, to meet the need of the sorting system, which is described in the following section. Fig. 2 illustrates the basic image processing and apple grading procedures. The fruit is segmented from the conveyor background by a two-class linear-discriminant classifier trained with RGB pixel intensities, followed by size and color assessments. The size is estimated by calculating the maximum equatorial diameter based on the detection of the fruit stem-calyx orientation (Fig. 2). A detailed description of fruit size estimation is given in Mizushima and Lu (2011a). The fruit color is determined based on the hue component after the acquired RGB image is converted into HSV (hue, saturation, value) format. Two predefined thresholds ($t_1 = 15$ and $t_2 = 30$) are applied to the hue image to classify the image pixels into three color classes, i.e., red (hue$<t_1$), stripe ($t_1 \leq$hue$\leq t_2$) and green (else). Users may also set different color and size grading criteria through the GUI of the program. The color grading is performed based on the calculated area percentages of these colors, while taking into account fruit varietal differences in skin colors (i.e., red and green) (Fig. 2 right). The camera is operated at a rate of 15 frames s^{-1} with the exposure time being adjustable by manually selecting appropriate parameters with the camera software, depending on whether apples are of the light (e.g., 'Golden Delicious') or dark (e.g., 'Red Delicious') colored variety. The actual imaging settings were selected and validated prior to the experiment for each variety of fruit. For each apple, the quality grade is determined by taking the average of the size and color metrics of the fruit in a sequence of 5–8 images depending on the actual conveyor speed. Results from a separate study (unpublished) on 'Golden Delicious' and 'Red Delicious' apples showed that when the grading and sorting system was running at a throughput of 9 apples s^{-1}, the images taken by the system were able to cover the entire surface area of each apple more than twice, regardless of fruit variety and size. Furthermore, the rotation pattern of apples was less affected by fruit shape (or variety); small apples generally rotated faster than bigger apples in completing a full, 360-degreen rotation. Hence, with the three test throughputs of 7.5–10.5 fruit s^{-1}, the imaging system was expected to cover the entire surface of each fruit at least twice.

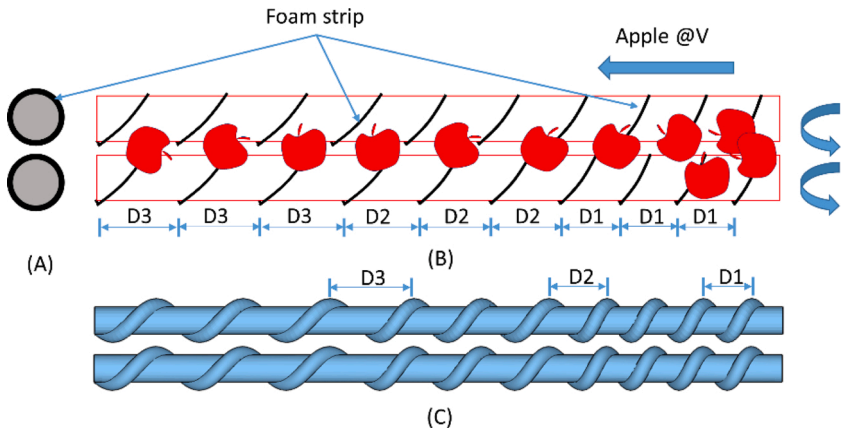


Fig. 1. One set of pitch-variable screw conveyors for apple singulation, rotation, and transportation: (A) cross-sectional view, (B) top view, and (C) 3-dimensional model. The pitch of the screw conveyors along the apple moving direction increases from D1 (83 mm) to D2 (102 mm) to D3 (121 mm).

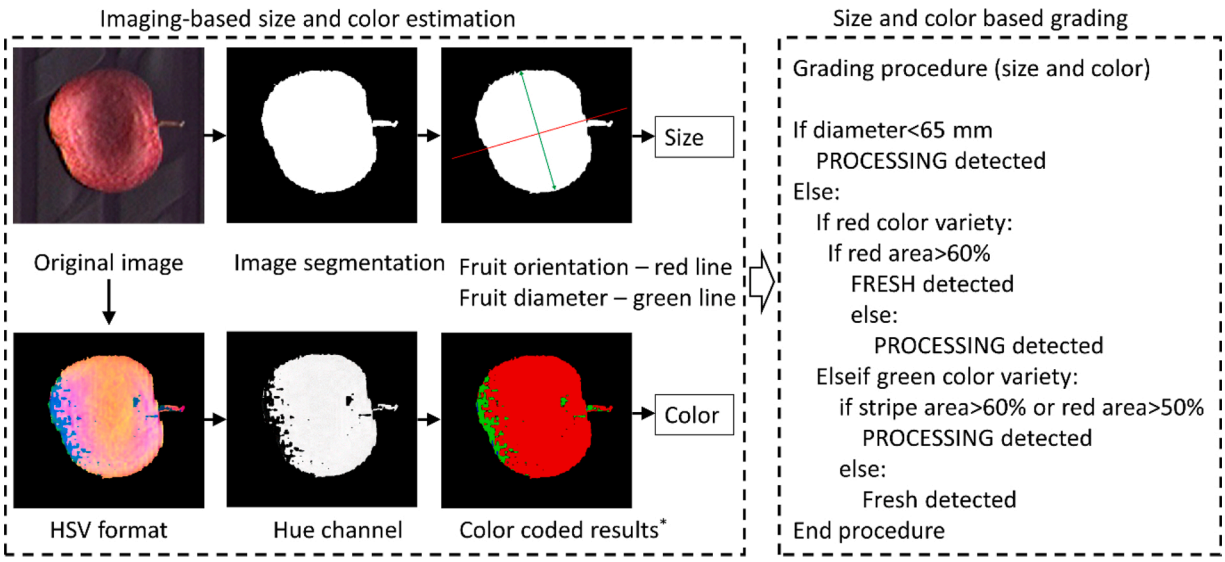


Fig. 2. Procedures of imaging-based apple grading (fresh and processing) according to size and color. *Hue channel is used for color evaluation: under the default settings, pixel values of < 15 are graded as red (coded in red color), $15 \leq$ pixel values ≤ 30 as green (coded in green color), and pixel values > 30 as stripe (coded in blue color). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

In the fruit size detection algorithm, when a stem is attached to the apple, it would be detected as part of the fruit using an approach of support vector machine and Ostu's method (Mizushima and Lu, 2013b). The detected stem, along with the calyx end, is used to assist in determining the fruit size. When the stem is absent from the fruit, the algorithm would identify the stem and calyx ends by analyzing the gradient changes in the contour of the segmented apple. The maximum diameter of the equational section of the fruit is determined to be perpendicular to the detected stem/calyx end axis. The size estimation algorithm has been able to estimate the maximum diameter of fruit of different sizes and varieties within an error of ±1.8 mm (Mizushima and Lu, 2013b). In the color analysis, the presence of the stem overall has a negligible effect of the color classification result, as reported in previous research (Mizushima and Lu, 2013b).

2.2. Sorting systems

2.2.1. Paddle sorter

After evaluation of different sorting designs including the cup-based and rotary sorters, we proposed and fabricated a new, simple, and high-throughput paddle sorter for sorting apples into two quality grades (processing and fresh). This sorter consists of a rotary solenoid with 12 V DC power supply, a rotational stroke of 40°, and a torque of 0.11 N m (McMaster-Carr, Aurora, OH, USA) and an aluminum paddle (75 mm length by 51 mm width), which is covered with 3-mm thickness foam (McMaster-Carr, Aurora, OH, USA) to avoid apple bruising when it is activated to change the movement direction of apples. If the apple exiting the screw conveyors is graded as fresh, the paddle sorter remains in open mode (Fig. 3C), so that it does not interfere the apple movement direction, and the apple directly goes to the fresh bin via a short inclined channel. If the apple is for processing, the paddle sorter will be activated (or in close mode, Fig. 3D) to push the apple sideways to the neighboring inclined channel, which leads to the processing bin.

This paddle sorter design has several advantages, compared to the cup-based and rotary sorters described earlier. Firstly, apples that are graded as fresh, after leaving the conveyors, directly go to the inclined channel (Fig. 3C), without touching the paddle that is in open mode. This would eliminate potential extra bruising to the fresh quality apples during the sorting process. Secondly, for processing-grade apples, only minimum force is needed to push the forward-moving fruit sideways to

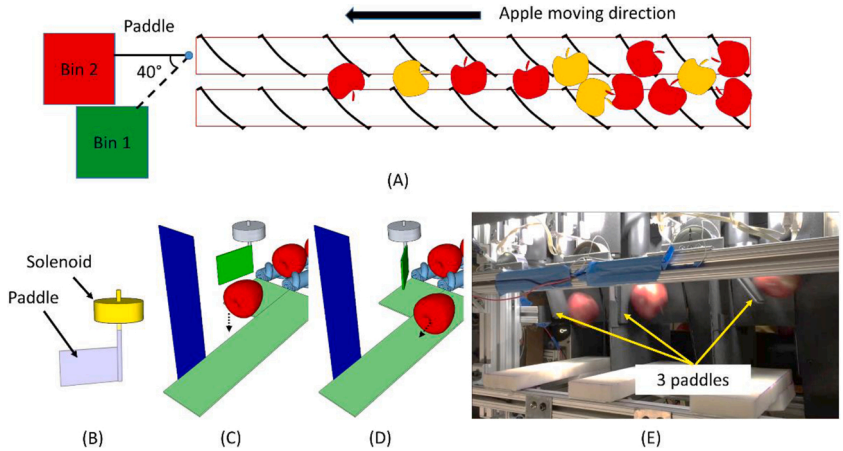


Fig. 3. Paddle sorter coupled with pitch-variable screw conveyors: (A) top view of the paddle sorter with screw conveyor, (B) paddle sorter closeup, (C) paddle in open mode for fresh-grade apples, (D) paddle in close mode for processing-grade apples, and (E) picture (from top) of the actual paddle sorters in working status – one paddle opens (center lane) and two paddles close (side lanes).

the neighboring channel (Fig. 3D). Thus, no bruising damage would be expected when the solenoid is being activated. Thirdly, the design is simple and compact, as the overall height of the sorter is only 111 mm (60 mm and 51 mm for the rotary solenoid and paddle, respectively). Finally, the sorter is driven by a 12 DC power supply, thus eliminating a hydraulic/mechanical power transmission system that is required for driving the cup-based and rotary sorters described earlier and for synchronization with the screw conveyors.

2.2.2. Sorter activation

Immediately after the imaging system has made grading decisions for the apples (Fig. 4), it sends signals to the solenoids to activate (or not activate) the sorter for each sorting lane. Accurate timing of the sorter activation is critical to ensure the graded fruit go to the right destinations (i.e., fresh and processing bins). The timing of the sorter activation is estimated based on the fruit transport speed (v) and the distance (Δx) between the fruit and paddle. Since apples are transported by the screw conveyors driven by a hydraulic motor (not shown in Fig. 4), it is reasonable to assume that the fruit transport speed (v) is the same as the linear translation speed of the conveyor, which is determined by the pitch of the screw conveyors and controlled by an adjustable variable flow control valve (Summit Hydraulics, Phoenix, AZ, USA) coupled with an in-line flow control valve (Kaman Fluid Power, Akron, OH, USA). A gear-tooth speed sensor (Cherry Americas, LLC, Kenosha, WI, USA) was used to monitor the real-time apple transport speed. The distance (Δx) from the sorter is estimated based on the fruit centroid position that is tracked in real time during image processing and fruit grading. Thus, the time (Δt) for the apple to reach the sorter is determined by $\Delta x/v$, which is used for triggering the sorter actions. As illustrated in Fig. 4, the imaging area (covered by the imaging chamber) was partitioned into three different subregions (designated as A, B, and C in Fig. 4), corresponding to three different states of apples, for facilitating apple tracking. The fruit grading process (Fig. 2) is being done in subregion A. By the time when an apple arrives at subregion B, its grade has been assigned. When the apple fully enters subregion B, its centroid is detected, based on which the distance between the apple and paddle is obtained (Δx in Fig. 4). If the apple is graded as fresh, the paddle will not be triggered (Fig. 3C); otherwise, the paddle would be triggered to close (Fig. 3D) after the calculated time has elapsed. The sorter activation is achieved via a digital acquisition device (USB-6009, National Instruments, Austin, TX, USA) connected to a relay (Vetco Electronics, Bellevue, WA, USA) that directly controls the paddle action.

2.3. Integrated apple infield grading and sorting system

After the separate modular grading and sorting systems have been constructed, they were then integrated into the whole system, including the base and imaging chamber. The base consists of the SRTM and paddle sorter, with the dimensions of 2400 × 350 × 610 mm (L × H × W); it is compact and convenient for transportation. The whole system was finally incorporated with a specially designed self-propelled harvest-assist machine, which contains automatic bin filling and handling systems as well as adjustable platforms at two heights (Lu et al., 2021). The image chamber with a trapezoid cross section and 610 × 610 mm (L × W) for the bottom and 610 mm for height, was designed and fabricated in modular concept for easy installation, and it can be readily installed on or removed from the base. A compact-size computer (Shuttle SH67H4, Intel Core i-7, 3.4 GHz, Windows 7 OS) is installed on the base for data processing, while a PCI-express adapter card (Fireboard Blue-e™ 1394a, Unibrain Inc., CA, USA) is used for interfacing the camera with the computer for real-time image collection.

2.4. Laboratory evaluation of the apple infield grading and sorting system

Previous research (Mizushima and Lu, 2013a, 2013b) has shown that the imaging system achieved superior performance in fruit size and color grading, and the SRTM provided satisfactory rotation functions from imaging (unpublished data). Hence, laboratory tests were focused on further evaluation of the system for the repeatability of grading fruit, sorting accuracy, and final fruit grading results due to bruise damage caused by the grading and sorting system. 'Red Delicious' (RD) and 'Golden Delicious' (GD) apples, manually picked from Michigan State University's Horticultural Teaching and Research Center (Holt, MI, USA) during the 2018 harvest season, were used for the laboratory tests. Harvested apples were kept in cold storage at 2 °C prior to the testing. Before each experiment, apples were removed from cold storage and kept at ambient temperature (~22 °C) for 4 h. For each of the three tests, the system throughputs were set at 7.5, 9.0, and 10.5 apples s^{-1}, which are equivalent to 2.5, 3.0, and 3.5 apples s^{-1} $lane^{-1}$, respectively.

2.4.1. Grading repeatability

Grading repeatability reflects the chances of an apple that would be graded into the same quality grade in multiple runs. Since there are three lanes (sets) of the pitch-variable screw conveyors, the repeatability test was conducted for intra- and inter-lanes. For the intra-lane

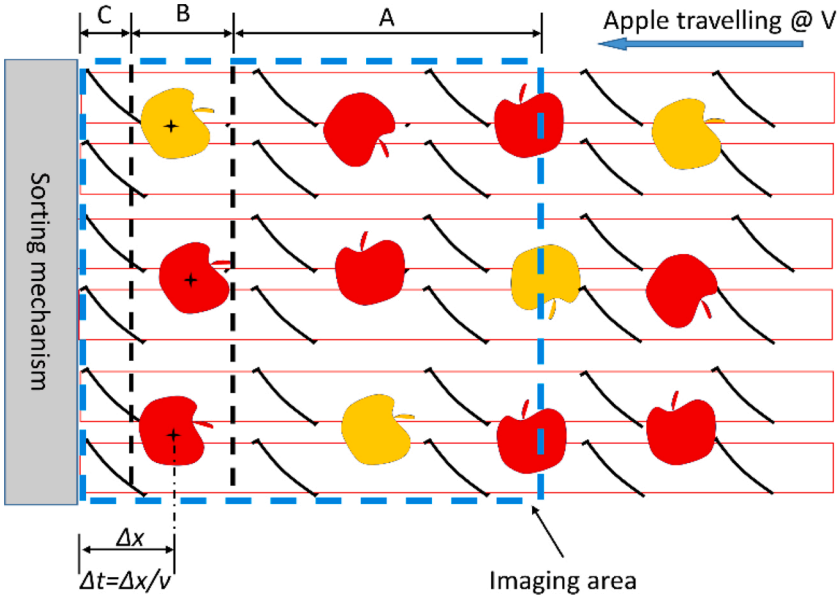


Fig. 4. Schematic of three imaging subregions for apple grading and tracking. In subregion A, apples are imaged and graded; in subregion B, the distance between the apple centroid and paddle and the time needed for apples to arrive at paddle are calculated; in subregion C, apples are no longer being tracked and analyzed. Δx, V and Δt represent the distance between apple and paddle, the apple's linear traveling speed, and the time gap before activating the paddle (only if the apple is graded as processing), respectively.

repeatability test, an apple went through the same lane twice to obtain two grading results. If the two results were the same, it was recorded as "1"; otherwise, it was recorded as "0". For the inter-lane repeatability test, one apple went through the three lanes one by one, and the three grading results were recorded and compared. If the three results were the same, it was recorded as "1"; otherwise, it was recorded as "0". The repeatability coefficient (RC) was calculated as the average of recorded results for all test apples; an RC value that was closer to "1" (or 100 %) would indicate that the grading system had a more reliable performance; otherwise, the repeatability is poor.

A total of 240 apples, 120 for each of the two varieties, were separated into eight groups (four groups per variety), with 30 apples for each group. For each variety, about 1/3 (40 apples) were processing-grade, while the other 2/3 (80 apples) were fresh market level. All the apples were first used for the intra-lane test and then for the inter-lane test. Each apple was used for nine times (six and three runs for intra- and inter-lane, respectively). During the test, a dedicated person stood at end of the screw conveyors to catch apples gently to avoid bruising.

2.4.2. Bruise evaluation

Bruise test was intended to evaluate if the system would cause mechanical damage to apples during the grading and sorting process at each of the three throughputs (i.e., 2.5, 3.0, and 3.5 fruit s^{-1} $lane^{-1}$). For each speed, 80 apples (40 per variety × 2 varieties) were used to run the experiment, with a total of 240 apples (80 apples per throughput × 3 throughputs). For each speed test, 40 apples of each variety (GD and RD) were randomly divided into four groups (10 apples per group), representing four replications for each variety at each speed setting. Prior to the experiment, each apple was visually inspected for pre-existing bruises, which were then marked with a permanent black marker. This procedure would avoid the pre-existing bruises from being considered as newly generated ones by the infield grading and sorting system.

Since the three lanes were exactly the same, only one lane (center one) was selected for the bruising evaluation test. Apples were manually fed onto the screw conveyor individually, and the paddle was set at the full close mode of 40° (Fig. 3D) during the entire test. Under the normal sorting process, the paddle sorter is activated right after a processing-grade fruit enters the sorting area. The paddle pushes or touches the incoming fruit, when it is still at an early stage of closing with a small angle. The smaller the touching angle, the less the impact on the moving apple. Hence, it is expected that when the paddle is at the full close position (40°), the impact force acting on the moving fruit would be greater than that at a smaller angle (10°~40°). During the test, a dedicated person stood at end of the sorter to receive apples exiting the sorter to avoid potential further bruising. After the test, these apples were kept at room temperature for 24 h and then visually inspected for new bruises, in terms of their number and total area. Based on the USDA standards (Peterson et al., 2010; USDA, 2019), fresh apples can be graded into three quality grades, i.e., Extra Fancy, Fancy, and Downgraded, the first two of which are considered fresh market quality and the last one is only suitable for processing. Within the Extra Fancy grade, there are four classes based on the degree of bruises, and there are two classes for Downgraded. In this study, we only graded the tested apples into three quality grades, based on the USDA fresh market apple standards as shown in Table 1.

Table 1
USDA classification of apple bruise damage (Peterson et al., 2010; USDA, 2019).

Quality Grade	Bruise Specification
Extra Fancy	No bruise, bruise diameter less than or equal to 12.7 mm, or total area of bruises < 127 mm^2
Fancy	Bruise diameter 12.7–19 mm or total area of multiple bruises < 283 mm^2
Downgraded	Bruises larger than the tolerances in "Fancy", or exhibiting cuts or punctures of any size

2.4.3. Sorting accuracy

Sorting accuracy is defined as the consistency between the imaging-based grading result and the apple destination. It measures the overall performance or coordination of the sorter and the imaging system in controlling the sorter activation. In the ideal condition, if an apple is graded as fresh, it is expected to arrive at the fresh bin; if graded as processing, it is expected to arrive at the processing bin. However, the sorting accuracy of the system can be affected by such factors as sorting speed (or throughput), the actual timing of activating the sorter for processing-grade apples, the mounting position of the paddle relative to the conveyors, and fruit condition (i.e., fruit variety and percent of processing-grade apples). Extensive preliminary tests were conducted in adjusting the sorter and the computer algorithm to ensure the system would perform at its optimum. To evaluate the system performance under different fruit load conditions, the sorting accuracy experiment was conducted in two extreme fruit load conditions: single (minimum load) and continuous (maximum load). In the single mode, one apple was fed to the screw conveyor and after this apple arrived at the bin, the next apple was then fed into the system. Both grading and destination results of individual apples were recorded. In the continuous mode, all apples were numbered at the stem area, so that these marks on the apples would not affect the fruit grading by the imaging system. Ten apples were first lined up on the entry section of the screw conveyor at maximum loading (i.e., one by one without a gap between apples), and the system was then turned on. Immediately after the system was on, a dedicated person standing by the system started to feed the lane to ensure that there was no gap between two neighboring apples, as there were 20 apples for each replication. These apples were finally collected at the two bins, with apple number and destination information recorded. Individual apples' grading results were automatically recorded by the computer, and then compared with the destination information. If an apple graded as fresh and arrived at Bin 2, or an apple graded as processing and arrived at Bin 1 (Fig. 3), it was recorded as "1". If an apple graded as fresh but arrived at Bin 1, or graded as processing but arrived at Bin 2, it was recorded as "0". Sorting accuracy coefficient (SAC) was calculated to be the average of all the results; the closer the SAC to "1" (or 100 %), the better performance of the system. Since the three lanes are the same in design, only one side lane was selected for experiment to simplify the fruit feeding and collecting as the side lane was easier to access during the experiment.

For the single mode test, 480 apples (80 apples per variety × 2 varieties × 3 throughputs) were used, and the same number of apples were used for the continuous mode test. For each sorting throughput, the 80 apples of each variety were randomly divided into four groups of 20 apples, each of which represented a replication. Thus, there were four replications for each test setting.

3. Results and discussion

3.1. Imaging system's grading repeatability

3.1.1. Intra-lane grading repeatability

The intra-lane grading RCs for the three throughputs of 2.5, 3.0, and 3.5 apples s^{-1} $lane^{-1}$ ranged from 92 % to 94 % and 90 % to 91 % for RD and GD, respectively (Fig. 5A). High RCs (>90 %) demonstrated the satisfactory intra-lane repeatability of the imaging-based grading system for both varieties. For each variety, RCs at the three throughputs were not significantly different from each other (same letters in each group), indicating that throughput did not affect the grading repeatability significantly. This demonstrated the performance robustness of the imaging system at different throughputs. Further analysis showed that under each of the three throughputs, RCs were not significantly different between GD and RD (Fig. 5B). This suggests that fruit variety or shape/color appear to have had little or no effect on the performance of the

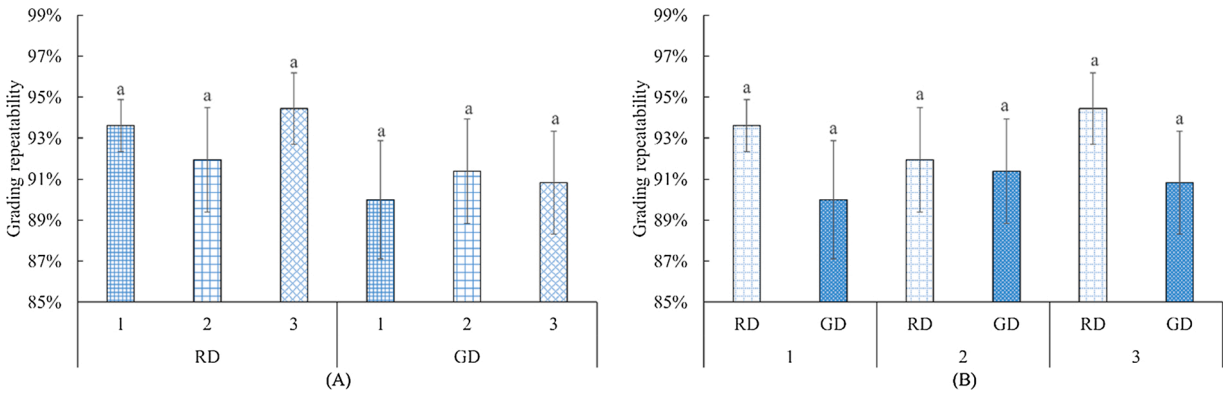


Fig. 5. Intra-lane grading repeatability results in terms of variety (A) and sorting speed (B). The bar and interval stand for the mean and standard deviation for 4 repetitions, respectively. In the figure, the numbers of 1, 2, and 3 represents the sorting speeds of 2.5, 3.0, and 3.5 apples s^{-1} $lane^{-1}$; RD and GD represent 'Red Delicious' and 'Golden Delicious' apples, respectively. Bars for each group with the same letters are not significantly different by Tukey test ($\alpha = 0.05$). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

imaging system, since GD ad RD apples are of different shape and distinct in the skin color; the former are of yellow color and more round, while the latter are red and elongated with irregular shape.

3.1.2. Inter-lane grading repeatability

Experimental results (Fig. 6) showed that RCs for inter-lane tests ranged from 88 % to 93 % and 81 % to 85 % for RD and GD, respectively. For each variety, throughput did not have a significant effect on the RC (Fig. 6A), which is the same as that for the intra-lane evaluation. For RD, the intra-lane RCs ranged from 92 % to 94 %, but the inter-lane RCs were lower ranging from 88 % to 93 %. The differences could be possibly because the fruit size correction algorithm did not result in consistent size estimations for the same apples that ran on the side lane versus on the center lane, as the viewing angles between the center and side lanes are different. The same trend occurred for GD apples, in which the lowest RC was 90 % for intra-lane, while the highest RC for inter-lane was 85 %. By re-grouping the results according to different sorting speeds (Fig. 6B), it was found that for the throughputs of 2.5 and 3.5 apples s^{-1} $lane^{-1}$, RD had a significantly higher RC over GD. It was observed during the experiment that some of the tested GD apples had very long and large stems. It is possible that the long and large stem protruding from the fruit body region may have been incorrectly identified as the fruit (not stem), thus significantly affecting the final grading results (both color and shape evaluation).

3.2. Apple bruise

Apple bruise is critical to determining whether the technology is suitable for commercial use. The industrial requirement is that at least 95 % apples be graded as Extra Fancy after sorting (Peterson et al., 2010). In other words, bruising caused by the grading and sorting system should not result in more than 5 % of the apples being graded below Extra Fancy. Results showed that for all three throughputs (i.e., 7.5, 9.0, and 10.5 apple s^{-1}), 100 % of the tested apples of both varieties were graded Extra Fancy. This clearly demonstrated that the grading and sorting system has exceeded the industrial bruise requirement. In addition, 58 % of RD apples and 55 % of GD apples under the three sorting speeds did not have any bruise after the test. Clearly, the use of soft and resilient materials (see details in the previous sections) for the screw conveyors, coupled with the new sorter design, was highly effective in reducing or eliminating bruising to apples, even when the system was running at high throughput. It should be mentioned that during the test, the paddle sorter was set in the close position, which represents the worst situation where the sorter could cause bruising damage to the graded apples. In practical situations, most of harvested apples are of the fresh market grade and therefore, the paddle sorter does not need to engage with these apples, thus eliminating the potential for bruising apples during sorting. The paddle sorter performed much better than other sorter designs in terms of fruit bruising, such as bucket- (Griffith, 1959) and cup-based (Sofu et al., 2016; Pothula et al., 2018) and rotary sorters (Lu et al., 2018). For these sorters, bruising is mostly

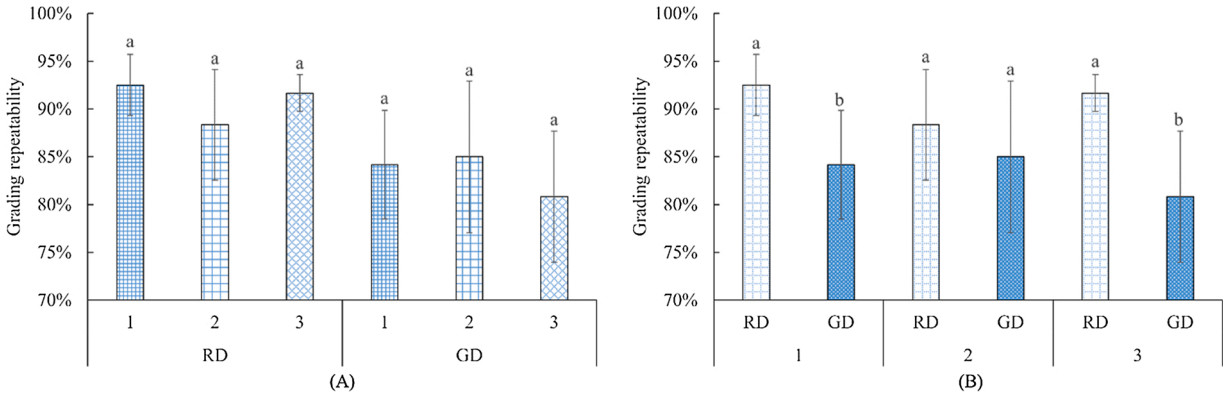


Fig. 6. Inter-lane grading repeatability in terms of variety (A) and throughputs (B). The bar and interval stand for the mean and standard deviation for 4 repetitions, respectively; 1, 2, and 3 represents 2.5, 3.0, and 3.5 apples s^{-1} $lane^{-1}$; RD and GD represents 'Red Delicious' and 'Golden Delicious', respectively. Bars for each group with different letters are significantly different by Tukey test ($\alpha = 0.05$). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

likely to occur when apples enter and exit the bucket, cup, or compartments when the system is operated at high throughput.

3.3. Sorting accuracy

Overall, the SACs under the 12 test settings (3 throughputs × 2 varieties × 2 feeding modes) are between 99.4 %, and 99.7 %, which demonstrated the superior performance of the system. These results also indicated that the paddle sorter has functioned well in accordance with signals from the imaging-based grading system, and the system accurately tracked, and calculated the time for, individual apples to arrive at the paddle. For the two fruit feeding modes (i.e., single and continuous) and two varieties of GD and RD apples, system throughput did not affect the sorting accuracy significantly, which indicates that the system was robust in the throughput range of 7.5–10.5 apples s^{-1}. Likewise, apple feeding mode also did not affect the SAC significantly. High SACs for high throughputs were achieved owing to accurate timing by the imaging system and the prompt response of the solenoid to activate the paddle. Preliminary tests showed that the paddle sorters could complete an open and close cycle within 100 ms, which is fast enough to meet the throughput requirement of 3.5 apples s^{-1} $lane^{-1}$ (~280 ms for a cycle). Though no other studies related to apple infield grading and sorting were reported, Soft et al. (2016) developed an apple grading and sorting system for inhouse use. Compared to their sorting accuracy of 79 %, our system performed significantly better (>99 %). In practical applications, apples may be fed into the system in a batch manner (i.e., apple-to-apple contact or overlapping) and all three lanes will be used. Hence, further evaluate of the sorter working under the actual orchard environment is needed.

4. Conclusions

An apple infield grading and sorting system, consisting of an imaging-based grading module, pitch-variable screw conveyors, and a paddle sorting mechanism, was developed. Laboratory tests were conducted for evaluating the system performance in terms of grading repeatability, fruit bruising, and sorting accuracy. The imaging-based grading system achieved above 90 % grading repeatability in the intra-lane test and above 81 % for the inter-lane test. Lower repeatability results for inter-lane tests suggest that further improvements to the imaging-based grading system, especially the image processing algorithm, are still needed for more consistent assessment of fruit color and size across the three lanes. The system has met or exceeded the industrial requirement for fruit bruising, with 100 % of the test apples being graded Extra Fancy, more than 55 % of which did not incur any bruise after grading and sorting. The paddle sorter achieved above 99 % sorting accuracies for both 'Red Delicious' and 'Golden Delicious' apples, when it was tested at the system throughputs of 7.5, 9.0, and 10.5 apples s^{-1} with two feeding modes. Overall, the system had robust performance and has met the requirements for apple infield grading and sorting, and thus it has potential for commercial application. However, further extensive field tests of the grading and sorting system are needed to evaluate its performance in fruit bruising and handling stems or limbs, leaves and dirt attached on the harvested fruit.

As a final note, the fruit grading and sorting system has been integrated with a specially designed self-propelled apple harvest-assist machine, which is also incorporated with automatic fruit bin filling and handling functions. The prototype machine was successfully tested in a commercial orchard during 2019 harvest season.

Disclaimer

Mention of commercial product names is for providing factual information only, and it does not imply endorsement of these products by USDA.

Declaration of Competing Interest

The authors report no declaration of interest.

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